

Binary classifier: $y = f(\vec{x}; \vec{w}) : \mathbb{R}^d \rightarrow y$

$$\vec{x} \in \mathbb{R}^d \\ y \in \{0, 1\}$$

Probabilistic Classifier: $f(\vec{x}, \vec{w}) : \mathbb{R}^d \rightarrow P(y|x)$

For Binary classification $P(y|x)$ is usually modeled as a Bernoulli distribution parametrized by $P(y=1|x) = p$
s.t. $P(y=0|x) = 1-p$

For Many classification, $P(y|x)$ is a Probability Mass function (PMF), i.e.

$$P(y|x) = [p_1 \ p_2 \ \dots \ p_n]$$

where $p_i \geq 0$ & $\sum_{i=1}^n p_i = 1$

Two Categories of Classifiers

1) Discriminative Classifiers

Classifiers that directly predict either y , given the input x , or predict $P(y|x)$ e.g. logistic regression models, where

$$P(y|x; w) = f(x; w) = \frac{1}{1 + e^{(-w^T x)}}$$

2) Generative Classifiers

These are classifiers that model the joint distribution $P(x, y)$ of the data and infer $P(y|x)$ using Bayes Rule:

$$P(y|x) = \frac{p(x|y) P(y)}{P(x)}$$

Here, $p(x|y)$: the class conditional data distribution
 $P(y)$: the a priori class probability.

Need for Generative Classifiers

Consider the example of digit recognition where we need to classify a binary image of a handwritten digit into two classes $\{5, 6\}$

$\therefore$ We have a binary classification problem. $y \in \{5, 6\}$

& our input is, say, an image of 30×30 binary pixels
i.e., $\vec{X} = \{x_1, x_2, \dots, x_n\}$ where $n = 900$ & $x_i \in \{0, 1\}$ $i = 1, \dots, n$

Directly estimating $P(y|\vec{X})$ may require answering unnatural questions like: given $\vec{X} = 0$; i.e. $x_1 = x_2 = \dots = x_{900} = 0$, what is the

PMF $P(y|\vec{X}=0)$

$$\text{i.e. } P(y=5|\vec{X}=0) = p$$

$$P(y=6|\vec{X}=0) = 1-p$$

} What should p be?
How do we estimate it?

In such cases, it makes sense to use a generative classifier where we estimate the posterior class probability

$$P(y|x) = \frac{P(x|y) \cdot P(y)}{P(x)} \quad \text{using Bayes Rule}$$

where $P(x|y)$ and $P(y)$ are as defined before.

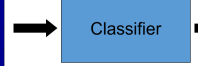
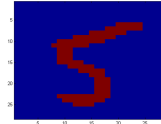
$$P(x) = \sum_{j \in Y} P(x|y=j) P(y=j)$$

is the probability distribution of the input data variable x , expressed as the marginalized distribution over the class variable y

We have seen examples where $P(x|y)$ is easier to estimate, e.g. in the Naive Bayes setting, where we assume conditional independence of all input variables, given the class

$$P(x_1, x_2, \dots, x_n | y) = \prod_{i=1}^N P(x_i | y)$$

• Digit Recognition



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• $x_1, \dots, x_n \in \{0, 1\}$ (Black vs. White pixels)

• $y \in \{5, 6\}$ (predict whether a digit is a 5 or a 6)

Often estimating $P(x_i|y)$ is substantially easier to estimate.

We had assumed that $x_i \in \{0, 1\}$. Now, let's relax this assumption to consider a more general setting.

GAUSSIAN DISCRIMINANT ANALYSIS (GDA)

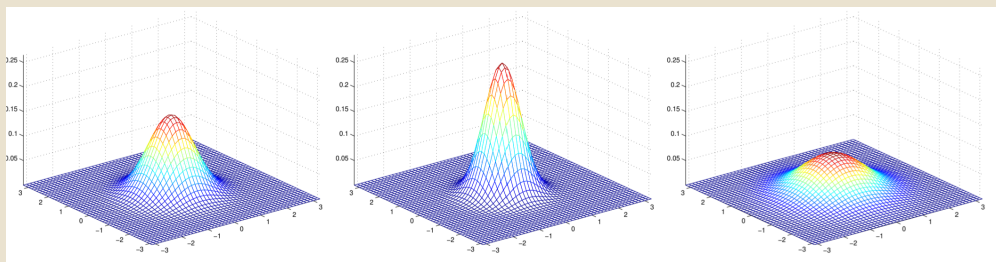
Recall that a multivariate Gaussian has the pdf

$$p_x(x; \mu, \Sigma) = \frac{1}{(\sqrt{2\pi})^d |\Sigma|^{1/2}} e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1} (x-\mu)}$$

Here $x, \mu \in \mathbb{R}^d$ and $\Sigma \in \mathbb{R}^{d \times d}$, $\Sigma \succeq 0$ are the parameters of the d -dimensional multivariate Gaussian distribution of the random variable $X \in \mathbb{R}^d$.
 (Notation for positive semidefiniteness.)

The mean $\mu = E[X] = \int x p_x(x; \mu, \Sigma) dx$

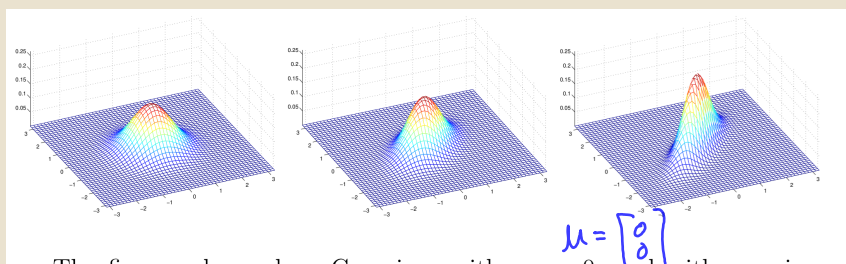
The covariance $\Sigma = \text{Cov}[X] = E[(X - E[X])(X - E[X])^T]$
 $= E[XX^T] - E[X]E[X]^T$



$$\mu = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \Sigma = I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\mu = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \Sigma = 0.1 I$$

$$\mu = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \Sigma = 2I$$



$$\mu = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

The figures above show Gaussians with mean 0, and with covariance matrices respectively

$$\Sigma = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}; \Sigma = \begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix}; \Sigma = \begin{bmatrix} 1 & 0.8 \\ 0.8 & 1 \end{bmatrix}.$$

Returning to our binary classification setting, say for class variable $y \in \{0, 1\}$, we have a model

$$P(X|y=0) = \mathcal{N}(\mu_0, \Sigma) \quad , \quad P(X|y=1) = \mathcal{N}(\mu_1, \Sigma)$$

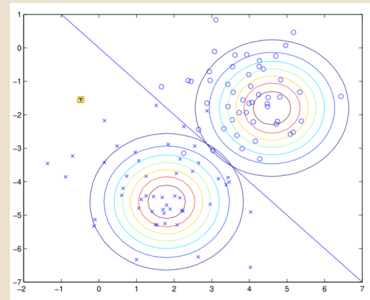
i.e., our class conditional data distribution is Gaussian, with mean μ_0 when $y=0$ and mean μ_1 when $y=1$

Since $y \in \{0, 1\}$ is binary, we can model it as a Bernoulli random variable, parametrized by ϕ , i.e.,

$$y \sim \text{Bernoulli}(\phi) \Rightarrow P(y) = \phi^y (1-\phi)^{(1-y)}$$

$$P(X=x|y=0) = \frac{1}{(\sqrt{2\pi})^d |\Sigma|^{1/2}} e^{-\frac{1}{2}(x-\mu_0)^T \Sigma^{-1} (x-\mu_0)}$$

$$P(X=x|y=1) = \frac{1}{(\sqrt{2\pi})^d |\Sigma|^{1/2}} e^{-\frac{1}{2}(x-\mu_1)^T \Sigma^{-1} (x-\mu_1)}$$



Now, the GDA model above has parameters $w = \{\phi, \mu_0, \mu_1, \Sigma\}$. Given a dataset $D = \{(x^1, y^1), (x^2, y^2), \dots, (x^N, y^N)\}$, the likelihood function using the joint distribution $p(x, y)$ is written as

$$\begin{aligned} \mathcal{L}(\phi, \mu_0, \mu_1, \Sigma) &= \prod_{i=1}^N p(x^i, y^i; \phi, \mu_0, \mu_1, \Sigma) \\ &= \prod_{i=1}^N p(x^i | y^i; \mu_0, \mu_1, \Sigma) p(y^i; \phi) \end{aligned}$$

Taking the log likelihood, we get

$$\log(\mathcal{L}(w)) = \sum_{i=1}^N \left[\log p(x^i | y^i; \mu_0, \mu_1, \Sigma) + \log p(y^i; \phi) \right] \quad \leftarrow$$

Introducing the binary indicator function

$$\mathbb{1}(z) = \begin{cases} 1 & \text{if } z \text{ is true} \\ 0 & \text{if } z \text{ is false} \end{cases}$$

Note that all these are disjoint cases that add up to N

$$L(w) =$$

$$\log(L(w)) = \sum_{i=1}^N \left[\mathbb{1}(y^i=1) \cdot \left[\log p(x^i | y^i=1; \mu_1, \Sigma) + \log p(y^i=1; \phi) \right] \right. \\ \left. + \mathbb{1}(y^i=0) \cdot \left[\log p(x^i | y^i=0; \mu_0, \Sigma) + \log p(y^i=0; \phi) \right] \right. \\ \left. + \mathbb{1}(y^i=2) \cdot \left[\log p(x^i | y^i=2; \mu_2, \Sigma) + \log p(y^i=2; \phi_2) \right] \right. \\ \left. + \dots \right. \\ \left. + \mathbb{1}(y^i=M) \cdot \left[\log p(x^i | y^i=M; \mu_M, \Sigma) + \log p(y^i=M; \phi_M) \right] \right]$$

if we had
more classes.

$$\left[\begin{array}{l} + \mathbb{1}(y^i=2) \cdot \left[\log p(x^i | y^i=2; \mu_2, \Sigma) + \log p(y^i=2; \phi_2) \right] \\ + \vdots \\ + \mathbb{1}(y^i=M) \cdot \left[\log p(x^i | y^i=M; \mu_M, \Sigma) + \log p(y^i=M; \phi_M) \right] \end{array} \right]$$

The decomposition above allows us to analyse the MLE estimate of the model parameters $w = \{\phi, \mu_0, \mu_1, \Sigma\}$

$$\frac{\partial L}{\partial \phi} = \frac{\partial}{\partial \phi} \left(\sum_{i=1}^N \mathbb{1}(y^i=1) \log p(y^i=1; \phi) + \mathbb{1}(y^i=0) \log p(y^i=0; \phi) \right)$$

For the Bernoulli distribution, $p(y^i=1; \phi) = \phi$ & $p(y^i=0; \phi) = 1 - \phi$

$$\frac{\partial L}{\partial \phi} = \frac{\partial}{\partial \phi} \left(\sum_{i=1}^N \mathbb{1}(y^i=1) \cdot \log \phi + \mathbb{1}(y^i=0) \log (1 - \phi) \right) \\ = \sum_{i=1}^N \mathbb{1}(y^i=1) \cdot \frac{1}{\phi} + \sum_{i=1}^N \mathbb{1}(y^i=0) \cdot \frac{1}{1 - \phi} \quad (-1)$$

$$= \frac{N_1}{\phi} - \frac{(N - N_1)}{1 - \phi} \\ = \frac{N_1 - N_1 \phi - N \phi + N \phi}{\phi(1 - \phi)}$$

$$N_1 = \sum_{i=1}^N \mathbb{1}(y^i=1)$$

Equating $\frac{\partial L}{\partial \phi} = 0$, we get $N_1 - N\phi = 0 \Rightarrow \phi = \frac{N_1}{N}$

$$\hat{\phi}_{MLE} = \hat{P}(y=1) = \frac{N_1}{N} = \frac{1}{N} \sum_{i=1}^N \mathbb{1}(y^i=1)$$

$$\hat{P}(y=0) = 1 - \hat{\phi}_{MLE} = 1 - \frac{N_1}{N} = \frac{1}{N} \sum_{i=1}^N \mathbb{1}(y^i=0)$$

Similar arguments can be made for ML estimates of prior probabilities $P(y)$ in an M -class classification problem.

Just as we obtained the MLE estimate of ϕ , we can use $\frac{\partial L}{\partial \mu_1}$ to obtain the MLE estimate of μ_1

$$\begin{aligned}
 \frac{\partial L}{\partial \mu_1} &= \frac{\partial}{\partial \mu_1} \left(\sum_{i=1}^N \mathbb{1}(y^i=1) \log p(x^i | y^i; \mu_1, \Sigma) \right) \\
 &= \frac{\partial}{\partial \mu_1} \left(\sum_{i=1}^N \mathbb{1}(y^i=1) \log e^{-\frac{1}{2}(x^i - \mu_1)^T \Sigma^{-1} (x^i - \mu_1)} + \text{const.} \right) \\
 &= \frac{\partial}{\partial \mu_1} \left(\sum_{i=1}^N \mathbb{1}(y^i=1) \left(-\frac{1}{2} (x^i - \mu_1)^T \Sigma^{-1} (x^i - \mu_1) \right) \right) + 0 \\
 &= \cancel{\frac{1}{2}} \sum_{i=1}^N \mathbb{1}(y^i=1) \left(\cancel{2} \cdot \Sigma^{-1} (x^i - \mu_1) \cancel{(-1)} \right) \\
 &= \Sigma^{-1} \left(\sum_{i=1}^N \mathbb{1}(y^i=1) (x^i - \mu_1) \right)
 \end{aligned}$$

Setting $\frac{\partial L}{\partial \mu_1} = 0$, we get

$$\sum_{i=1}^N \mathbb{1}(y^i=1) x^i - \sum_{i=1}^N \mathbb{1}(y^i=1) \mu_1 = 0$$

$$\Rightarrow \hat{\mu}_{1, \text{MLE}} = \frac{\sum_{i=1}^N \mathbb{1}(y^i=1) x^i}{\sum_{i=1}^N \mathbb{1}(y^i=1)} = \frac{1}{N_1} \sum_{\substack{i: y^i=1 \\ i=1:N}} x^i$$

Similarly $\hat{\mu}_{0, \text{MLE}}$ and $\hat{\Sigma}_{\text{MLE}}$ can be derived.

$$\hat{\mu}_{0, \text{MLE}} = \frac{\sum_{i=1}^N \mathbb{1}(y^i=0) x^i}{\sum_{i=1}^N \mathbb{1}(y^i=0)}$$

$$\hat{\Sigma}_{\text{MLE}} = \frac{1}{N} \sum_{i=1}^N (x^i - \mu_{y^i}) (x^i - \mu_{y^i})^T$$

This indexing indicates the covariance is computed classwise.

Had we assumed different Σ_0 & Σ_1 for $y=0$ & $y=1$ resp, the resulting ML estimates of Σ_j would have been

$$\hat{\Sigma}_j_{ML} = \frac{\sum_{i=1}^N \mathbb{1}(y^i=j) (x^i - \mu_j)(x^i - \mu_j)^T}{\sum_{i=1}^N \mathbb{1}(y^i=j)} \quad j \in \{0,1\}$$

For our 2-class setting, with each class being modeled as a Gaussian distribution, the joint likelihood for our dataset $D = \{(x^i, y^i)\}_{i=1}^N$, $x^i \in \mathbb{R}^d$, $y^i \in \{0,1\}$ is given as

$$\begin{aligned} \ell(\phi, \mu_0, \mu_1, \Sigma) &= \prod_{i=1}^N p(x^i, y^i; \phi, \mu_0, \mu_1, \Sigma) \\ &= \prod_{i=1}^N p(x^i | y^i; \mu_0, \mu_1, \Sigma) \cdot p(y^i; \phi) \end{aligned}$$

Now, consider the setting where from our dataset $D = \{(x^i, y^i)\}_{i=1}^N$, only a subset $D_X \subset D$ is given.

$$D_X = \{x^i\}_{i=1}^N \quad x^i \in \mathbb{R}^d$$

Then we can only write the joint likelihood in terms of $P(x)$

$$\ell(\omega) = \prod_{i=1}^N p(x^i; \omega)$$

$\therefore$ the log likelihood is

$$L(\omega) = \sum_{i=1}^N \log p(x^i; \omega)$$

$$= \sum_{i=1}^N \sum_{\substack{y^i \in \{0,1\} \\ z^i}} \log p(x^i | y^i; \mu_0, \mu_1, \Sigma) p(y^i; \phi)$$

only if y^i were known!

